

Quantum Key Distribution

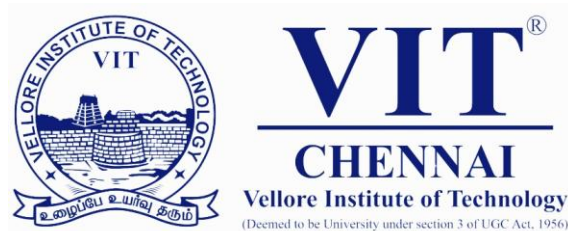
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LIST OF ACRONYMS

Acronym	Full Form
QKD	Quantum Key Distribution
BB84	Bennett-Brassard 1984 Protocol
QBER	Quantum Bit Error Rate
E91	Ekert 1991 Protocol
B92	Bennett 1992 Protocol
MDI-QKD	Measurement Device Independent Quantum Key Distribution
SDK	Software Development Kit
IBM Q	IBM Quantum
API	Application Programming Interface
UI/UX	User Interface / User Experience
CSV	Comma-Separated Values
JSON	JavaScript Object Notation
RSA	Rivest–Shamir–Adleman Encryption Algorithm
AES	Advanced Encryption Standard
ECC	Elliptic Curve Cryptography
PNS	Photon Number Splitting Attack
WebGL	Web Graphics Library
HTML	HyperText Markup Language
CSS	Cascading Style Sheets

INTRODUCTION

Overview

In today's digital era, secure communication has become one of the most crucial aspects of information technology. Traditional cryptographic techniques, such as RSA and AES, rely on mathematical algorithms that assume the computational difficulty of certain problems. However, with the rapid advancement of computing power and the emergence of quantum computers, these classical cryptographic systems are becoming increasingly vulnerable. Quantum computers have the potential to break many existing encryption methods by efficiently solving problems that are currently considered intractable for classical systems.

To address this growing threat, researchers have turned to **quantum cryptography**, a revolutionary field that leverages the principles of quantum mechanics to achieve secure communication. Unlike classical cryptographic methods, quantum cryptography offers *unconditional security*, meaning that its safety is not based on computational hardness but on the fundamental laws of physics. One of the most promising and widely studied applications of quantum cryptography is **Quantum Key Distribution (QKD)**.

What is Quantum Key Distribution (QKD)?

Quantum Key Distribution is a technique that enables two parties—commonly referred to as **Alice** (the sender) and **Bob** (the receiver)—to generate and share a secret encryption key through a quantum channel. This key can then be used for secure communication over classical networks. The unique property of QKD lies in the fact that any attempt by an eavesdropper (**Eve**) to intercept the key will disturb the quantum states being transmitted, alerting the legitimate users to the presence of an intrusion.

QKD uses individual photons as quantum bits or **qubits**, which can exist in a superposition of states. The behavior of these qubits is governed by two fundamental principles of quantum mechanics:

1. **Heisenberg's Uncertainty Principle** – it is impossible to measure a quantum state without disturbing it.
2. **Quantum No-Cloning Theorem** – it is impossible to make an exact copy of an unknown quantum state.

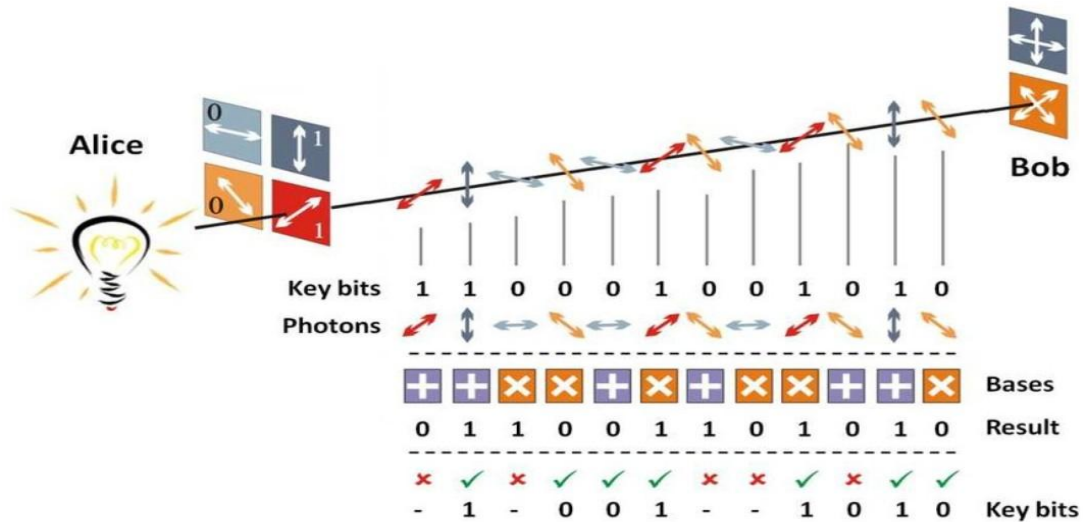
Because of these principles, any eavesdropping attempt changes the quantum states in a detectable way, ensuring that the key exchange remains secure.

The BB84 Protocol

The **BB84 protocol**, proposed by **Charles Bennett and Gilles Brassard in 1984**, was the first practical implementation of QKD. It uses two sets of polarization bases—rectilinear (0° , 90°) and diagonal (45° , 135°)—to encode and transmit bits.

- Alice randomly chooses a bit value (0 or 1) and a polarization basis for each photon she sends.
- Bob, on the receiving end, randomly chooses a basis to measure each photon.
- After transmission, Alice and Bob publicly compare the bases used for each photon (without revealing the bit values).
- They retain only those bits where their bases matched, forming a shared secret key.

If an eavesdropper tries to intercept the photons, their measurement will disturb the states, introducing detectable errors in the final key sequence. This process allows Alice and Bob to verify the integrity of the key before using it for secure communication.



Need for Quantum Key Distribution

The increasing power of quantum computing threatens the security of existing cryptographic algorithms, such as RSA and ECC, which depend on the difficulty of factoring large numbers or solving discrete logarithmic problems. Quantum algorithms like **Shor's algorithm** can potentially break these encryption schemes in polynomial time, rendering them obsolete.

In contrast, QKD offers a **future-proof solution** because it is based on physical properties rather than mathematical assumptions. This makes it an ideal candidate for next-generation secure communication systems, including financial networks, defense communications, and healthcare data transfer.

Project Objective

The main objective of this project is to **simulate and analyze the BB84 Quantum Key Distribution protocol** using **Python and IBM's Qiskit library**. The project aims to:

- Understand the theoretical concepts behind QKD and BB84 protocol.
- Simulate and analyze the effect of eavesdropping and quantum noise on key generation.
- Evaluate the security and efficiency of the implemented model through experimental results.

Scope of the Project

This project focuses primarily on simulation rather than physical implementation. Using **IBM's Qiskit**, a powerful quantum computing SDK, we can model quantum circuits, measure qubit states, and simulate communication between sender and receiver. The simulation provides insights into how QKD operates under real-world conditions, including noise, measurement errors, and interception attempts. Future extensions of this work could include integrating the BB84-based QKD model with **post-quantum cryptographic protocols**, implementing it on **IBM Quantum hardware**, or testing hybrid systems that combine classical and quantum encryption for practical security applications.

LITERATURE SURVEY

Introduction

A literature survey is an essential component of any research project, as it provides a detailed overview of existing work related to the topic and highlights the gaps that motivate further exploration. This chapter presents a comprehensive study of the existing research on **Quantum Key Distribution (QKD)**, focusing on the **BB84 protocol** and its variants. It also reviews the advancements in **quantum communication simulations** using **IBM's Qiskit** and other quantum programming frameworks.

The evolution of QKD represents a paradigm shift in secure communication. Since the introduction of the BB84 protocol in 1984, numerous research studies and implementations have contributed to enhancing the efficiency, reliability, and practicality of quantum cryptography systems. This chapter organizes the literature into key areas: foundational concepts, protocol developments, experimental implementations, and simulation tools.

Evolution of Quantum Cryptography

The concept of quantum cryptography originated from the groundbreaking idea that the **laws of quantum mechanics** could be used to guarantee information security. In 1984, **Bennett and Brassard** introduced the **BB84 protocol**, marking the birth of quantum cryptography. The innovation lay in using **quantum bits (qubits)** instead of classical bits, enabling secure key exchange that is resistant to eavesdropping.

Following this development, **Ekert (1991)** proposed the **E91 protocol**, which relied on the phenomenon of **quantum entanglement** for key generation. Unlike BB84,

which uses single photons, E91 employs entangled photon pairs to ensure that any interference by an eavesdropper disturbs the correlation between the two particles, revealing the intrusion.

Subsequent research by **Bennett (1992)** introduced the **B92 protocol**, a simplified version of BB84 using only two non-orthogonal quantum states. While less efficient in key generation, it further demonstrated the practicality of QKD in real-world scenarios

Theoretical Foundations of QKD

The fundamental principles underlying QKD are derived from **quantum mechanics**, particularly the **Heisenberg Uncertainty Principle** and the **No-Cloning Theorem**. These principles ensure that:

- Measurement of a quantum system inevitably disturbs it.
- Unknown quantum states cannot be perfectly duplicated.

Heisenberg's Uncertainty Principle states that it is impossible to simultaneously determine both the position and momentum (or, more generally, certain pairs of physical properties) of a particle with absolute precision. In the context of QKD, this principle guarantees that any attempt to intercept or measure a photon changes its state, thereby alerting the communicating parties to the presence of an eavesdropper.

The Quantum No-Cloning Theorem, established by **Wootters and Zurek (1982)**, asserts that it is physically impossible to create an identical copy of an unknown quantum state. This means that an eavesdropper cannot simply clone qubits and forward them to the receiver without introducing detectable errors.

These foundational principles provide the **unconditional security** that distinguishes QKD from all forms of classical cryptography.

REVIEW OF QKD PROTOCOLS

❖ BB84 Protocol (1984)

Developed by **Bennett and Brassard**, the BB84 protocol is the first and most extensively studied quantum key distribution protocol. It uses four polarization states—two in the rectilinear basis (0° and 90°) and two in the diagonal basis (45° and 135°)—to encode bit values 0 and 1. The key advantage of BB84 lies in its simplicity and ability to detect eavesdropping through measurement disturbances. Numerous experimental implementations of BB84 have demonstrated secure communication over both optical fiber and free-space channels.

❖ E91 Protocol (1991)

Proposed by **Artur Ekert**, the E91 protocol relies on **entanglement-based key generation**. Entangled photon pairs are shared between Alice and Bob, and their correlated measurement results are used to establish a shared secret key. The security of this protocol is verified using **Bell's Inequality**, which ensures that any attempt at eavesdropping destroys quantum correlations.

❖ B92 Protocol (1992)

This simplified protocol, proposed by Bennett, uses only two non-orthogonal quantum states instead of four. It reduces the complexity of implementation but also lowers the key generation rate. However, it is still valuable for understanding the fundamental aspects of quantum communication.

❖ SARG04 and Decoy-State Protocols

The **SARG04 protocol**, introduced by Scarani et al., enhances the security of BB84 by using non-orthogonal state discrimination techniques, making it more resistant to **Photon Number Splitting (PNS)** attacks. The **Decoy-State BB84** protocol, introduced later, improves key generation rates and transmission distance by detecting photon-level eavesdropping through decoy photon pulses.

❖ Measurement Device Independent (MDI) QKD

The **MDI-QKD** protocol addresses the vulnerability of detector side-channel attacks by allowing both communicating parties to send quantum states to an untrusted third-party measurement device. This eliminates the need to trust the detection process, thereby enhancing the overall system security.

Quantum Communication and Qiskit Simulations

The practical realization of QKD has been facilitated by advances in quantum computing frameworks such as **IBM Qiskit**, **Cirq**, and **QuTiP**. Qiskit, in particular, provides an open-source SDK for designing and simulating quantum circuits, allowing students and researchers to test and visualize QKD protocols without needing physical quantum hardware.

Numerous studies have demonstrated the use of Qiskit to model quantum communication systems:

- **IBM Quantum Experience (2018–present)**: Provided a cloud-based platform for QKD simulation and testing of quantum circuits.
- **Pérez-Delgado et al. (2020)**: Showed that BB84 can be effectively implemented on IBM's 5-qubit quantum computer, demonstrating secure key generation and detection of eavesdropping.
- **Patel et al. (2022)**: Utilized Qiskit to simulate noise effects on quantum key exchange, analyzing bit error rates and decoherence impacts.

These studies confirm that Qiskit offers a reliable and accessible environment for understanding the behavior of quantum systems and cryptographic processes.

Challenges in Quantum Key Distribution

Despite its promise, several challenges hinder the large-scale implementation of QKD systems:

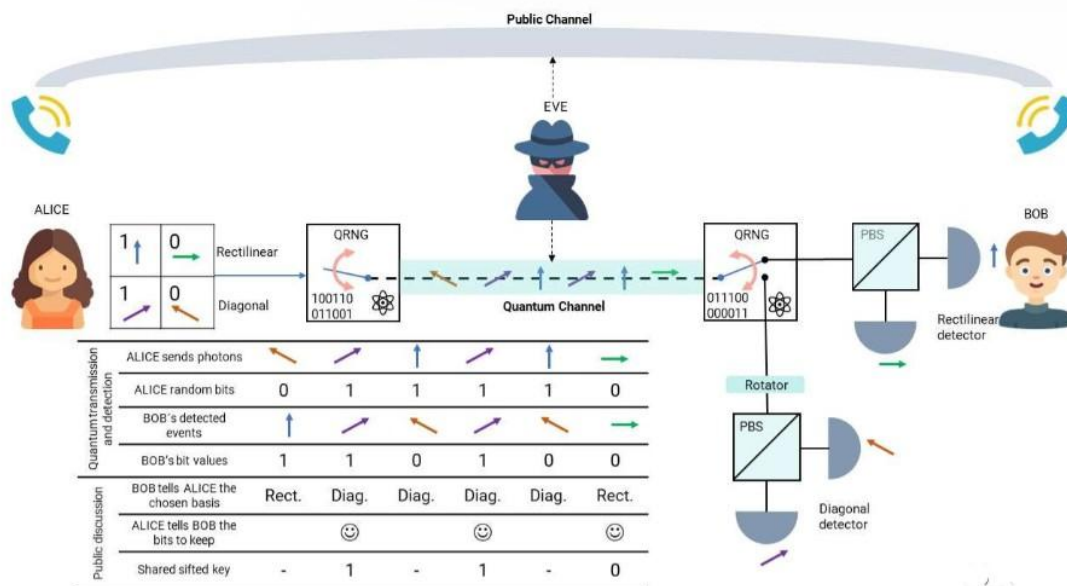
- **Photon Loss and Decoherence:** Transmission over long distances causes signal degradation, leading to lower key generation rates.
- **Hardware Limitations:** Quantum devices such as single-photon sources and detectors are still expensive and prone to errors.
- **Channel Noise:** External disturbances in optical fibers and free-space channels can introduce bit errors.
- **Scalability and Standardization:** There is a lack of unified standards and protocols for integrating QKD into existing infrastructures.

Researchers continue to explore **quantum repeaters**, **error correction codes**, and **hybrid encryption models** to overcome these limitations.

PROJECT FEATURES AND IMPLEMENTATION

Introduction

The **BB84 Quantum Key Distribution 3D Simulator** is an interactive web-based educational platform designed to visualize the working principles of **Quantum Key Distribution (QKD)**. This simulator was developed as part of an **Engineering Physics curriculum** to help students understand how quantum mechanics ensures secure communication between two parties. The project uses a combination of **interactive visualization, real-time data analytics, and educational theory** to explain the BB84 protocol in an engaging and intuitive manner.



Website Overview

The simulator provides users with a 3D environment where they can **visualize photon transmission, polarization, and basis selection**—the key processes of the BB84 protocol. It demonstrates how **Alice (the sender)** and **Bob (the receiver)** establish a shared secret key and how **Eve (the eavesdropper)** affects the communication channel.

The platform integrates both **learning modules** and **interactive simulation tools**, allowing users to:

- Understand the **theory behind BB84**.
- Observe **real-time quantum behavior** through 3D visualization.
- Analyze **key generation efficiency** and **security metrics**.

Core Features

Home Page Navigation

The home page provides intuitive navigation with the following main action buttons:

- **Start 3D Simulation:** Launches the BB84 protocol simulator.
- **Learn the Theory:** Opens educational resources explaining the working of BB84.
- **Meet the Team:** Displays project contributors and their roles.

Feature Highlights:

- Photon polarization visualization.
- Basis matching demonstration.
- Secure key extraction and display.

Theory Section

What is BB84?

The BB84 protocol, developed by **Charles Bennett and Gilles Brassard in 1984**, is the first and most widely studied **Quantum Key Distribution** method. It utilizes quantum states of photons to establish a **shared secret key** between two communicating parties while allowing them to **detect eavesdropping attempts**.

Photon Polarization Bases

Basis	Symbol	Bit 0	Bit 1
Rectilinear	(+)	Vertical () – Cyan	Horizontal (—) – Blue
Diagonal	(×)	Diagonal (/) – Purple	Anti-diagonal (\) – Pink

Protocol Steps

1. **Alice prepares photons:** Randomly selects bits and bases for encoding.
2. **Alice sends photons:** Transmits encoded photons through a quantum channel.
3. **Bob measures photons:** Randomly chooses measurement bases.
4. **Basis reconciliation:** Alice and Bob compare their bases (publicly) without revealing bit values.
5. **Error checking:** Subset verification to detect possible eavesdropping.

Secret key extraction: Bits with matching bases form the secure shared key.

Eavesdropping Detection

When **Eve** attempts to intercept photons, the **quantum states collapse** due to measurement, introducing random errors. Typically, a **50% error rate** is observed in mismatched bases, indicating the presence of eavesdropping. This unique property ensures that **any intrusion is immediately detectable**.

Interactive Simulation Controls

The simulator allows users to manipulate various parameters and visualize their effects in real-time.

Adjustable Parameters

- **Number of Photons:** Adjustable between 8 and 32 (default: 16).
- **Include Eavesdropper (Eve):** Toggle on/off.
- **Run Simulation:** Executes the BB84 protocol sequence.
- **Reset:** Clears all simulation data.

Usage Guide

1. Select the number of photons.
2. Toggle Eve to simulate eavesdropping.
3. Click “Run Simulation” to view real-time animation.

Color Indicators

- **Green Photons:** Basis match (kept as part of key).
- **Red Photons:** Basis mismatch (discarded).

3D Visualization Engine

The simulator’s visualization engine is built using **WebGL** and offers real-time rendering of quantum communication.

Visual Layout

- **Alice (Sender):** Blue 3D box (left side).
- **Bob (Receiver):** Purple 3D box (right side).
- **Photons:** Colored spheres representing quantum states.

Color Scheme

Color	Meaning
Green	Basis match (secure key bit)
Red/Pink	Basis mismatch (discarded)

Display Features

- Real-time photon transmission animation.
- Layered visualization for clarity.
- On-screen color legend for reference.

Analytics Dashboard

The dashboard provides **quantitative feedback** about the simulation's performance and security.

Key Metric Cards

- **Total Photons:** Example – 16
- **Bases Matched:** Example – 6–7
- **Key Length:** Example – 6–7 bits
- **Efficiency:** Example – 38–44%

Charts and Panels

- **Key Generation Efficiency:** Bar graph depicting usable photon percentage.
- **Basis Distribution:**
 - Alice's Bases: Rectilinear vs Diagonal.
 - Bob's Bases: Rectilinear vs Diagonal.

Security Assessment

- **Without Eve:**
✓ Secure Key Generated — *No errors detected. The key is safe for encryption.*
- **With Eve:**
⚠ Security Compromised — *Error rate of 14.3% detected. Possible eavesdropping by Eve.*

Eavesdropping Activity Panel

- Photons Intercepted: e.g., 9
- Error Rate: e.g., 14.3%
- Errors in Matched Bases: e.g., 1

Key Generation Results

The simulator displays the **final secret key** after the protocol run.

Generated Secret Key

- **Example Output:** 1 1 0 1 0 0
- **Key Length:** 6 bits
- **View Options:** Binary or numerical representation.

Export Options

- **Export as CSV:** Download data as spreadsheet.
- **Export as JSON:** Download structured key data.

Basis Reconciliation Table Example

#	Alice Bit	Alice Basis	Bob Basis	Polarization	Bob Bit	Result
1	0	+	×	/	1	✗ Discarded
2	1	×	×	\	1	✓ Kept

This table provides **complete transparency** for each photon's journey throughout the protocol.

User Interface and Design

Visual Aesthetics

- Dark, **space-themed background** with animated stars.
- Gradient **blue-to-purple color palette**.
- Clean, modern typography and card-based modular layout.

Navigation and UX

- Tab-based interface for Analytics and Results.
- Smooth transition animations for enhanced interactivity.

Technical Implementation Summary

Feature Category	Implementation Details
Simulation Engine	Real-time BB84 protocol execution (8–32 photons)
3D Visualization	WebGL rendering of photons, Alice, and Bob
Eavesdropping Simulation	Toggle-based Eve attack with error analysis
Data Analytics	Real-time metrics, efficiency charts, and visual panels
Educational Content	Theory section explaining BB84 and quantum basics
Data Export	Supports CSV and JSON output formats
Security Assessment	Automatic detection of compromised keys
Interactivity	Adjustable parameters and live updates

Educational Value

The **BB84 Quantum Key Distribution 3D Simulator** bridges the gap between **theoretical quantum concepts** and **practical understanding**.

It enables users to:

- Grasp the **fundamentals of QKD** intuitively.
- Visualize **basis reconciliation** and **photon behavior**.
- Understand **eavesdropping detection** through error analysis.
- Observe statistical efficiency (~50%) in realistic conditions.

This makes the tool a **valuable educational aid** in modern engineering physics and quantum computing courses.

Future Enhancements

The following improvements are proposed to extend the simulator's functionality and realism:

- Integration with **IBM Qiskit** for executing real quantum experiments.
- Addition of **multi-user mode** for real-time online communication.
- Implementation of **quantum noise modeling** for realistic channel simulation.
- Visualization of **Quantum Bit Error Rate (QBER)** with detailed analytics.
- **Mobile-friendly version** for enhanced accessibility.
- Exporting **visual reports as PDFs** for academic submissions.

Project Repository

GitHub Repository: https://github.com/PRODHOSH/bb84_simulation

Developed by:

Prodhosh V. S.

ALGORITHMS AND SOFTWARE USED

Introduction

This chapter discusses the **algorithmic logic** and **software framework** utilized in developing the **BB84 Quantum Key Distribution (QKD) simulation**.

The BB84 protocol was chosen due to its simplicity, efficiency, and proven reliability as the first practical quantum cryptographic system. The algorithm simulates the secure exchange of encryption keys between two users—**Alice** (sender) and **Bob** (receiver)—while detecting any eavesdropping attempts by a third party, **Eve**. To achieve realistic simulation and visualization, the project employs a structured algorithm implemented using **Python** and **IBM's Qiskit** library. These tools enable the modeling of **quantum bits (qubits)**, **measurement processes**, and **random basis selection**, which are essential for replicating quantum communication behavior.

Software Tools Used

Software / Technology	Purpose / Functionality
Python 3.x	Used as the primary programming language for implementing the BB84 algorithm.
IBM Qiskit	A Python-based SDK for quantum computing; used for simulating qubit preparation, measurement, and visualization.
NumPy / Random Library	Used for random generation of bits and bases (Z and X).
Matplotlib / Plotly	Used for visualizing photon states, efficiency, and error analysis.
WebGL (for 3D version)	Provides real-time visualization of photon transmission in browser-based simulation.
HTML / CSS / JavaScript	Used for front-end design and UI/UX elements of the simulator website.

Algorithm Overview

The BB84 simulation algorithm is designed to follow the **logical sequence of actual quantum key distribution**, ensuring realistic representation of photon preparation, transmission, measurement, and key sifting.

The algorithm comprises five main stages:

- 1. Alice's Preparation:**
Alice randomly selects bit values (0 or 1) and bases (Z or X) for each photon and prepares corresponding qubit states.
- 2. Eve's Interception:**
Eve randomly intercepts a fraction of the photons, measures them in random bases, and re-sends them—disturbing the original states (as per the Quantum No-Cloning Theorem).
- 3. Bob's Measurement:**
Bob receives the photons and measures each one in a randomly chosen basis. Depending on the match between Alice's and Bob's bases, measurement outcomes may or may not align.
- 4. Sifting (Key Agreement):**
Alice and Bob publicly compare their bases (not bits). They retain only those bits where the bases matched, forming the shared secret key.
- 5. Key Extraction and Output:**
The system displays all sequences—Alice's bits, bases, Bob's measurements, and the final shared key. The simulation also indicates whether eavesdropping occurred.

Pseudocode for BB84 Quantum Key Distribution

Below is the structured pseudocode for the BB84 simulation algorithm:

Pseudo Code – BB84 Quantum Key Distribution (QKD)
Simulation

```
BEGIN BB84_SIMULATION
```

```
    INPUT:  $n \leftarrow$  number of photons (between 8 and 32)
```

```
    IF  $n < 8$  OR  $n > 32$  THEN
```

```
         $n \leftarrow 8$ 
```

```
        PRINT "Defaulting to 8 photons"
```

```
    END IF
```

```
    // --- STEP 1: Alice's Preparation ---
```

```
    FOR  $i = 1$  TO  $n$  DO
```

```
         $\text{alice\_bits}[i] \leftarrow \text{RANDOM}(0 \text{ or } 1)$ 
```

```
         $\text{alice\_bases}[i] \leftarrow \text{RANDOM}(Z \text{ or } X)$ 
```

```
        PREPARE  $\text{qubit}[i]$  using (bit, basis)
```

```
    END FOR
```

```
    // --- STEP 2: Eve's Interception ---
```

```
     $\text{eve\_indices} \leftarrow \text{RANDOM sample of } (n / 4) \text{ photons}$ 
```

```
    FOR each index  $i$  in  $\text{eve\_indices}$  DO
```

```
         $\text{eve\_basis} \leftarrow \text{RANDOM}(Z \text{ or } X)$ 
```

```
        MEASURE  $\text{qubit}[i]$  in  $\text{eve\_basis}$ 
```

```
        DISTURB  $\text{qubit}[i]$  (collapse occurs)
```

```
    END FOR
```

```

// --- STEP 3: Bob's Measurement ---
FOR i = 1 TO n DO
    bob_bases[i] ← RANDOM(Z or X)
    MEASURE qubit[i] in bob_bases[i]
    bob_results[i] ← measured bit value
END FOR

// --- STEP 4: Sifting (Key Agreement) ---
shared_key ← []
FOR i = 1 TO n DO
    IF alice_bases[i] == bob_bases[i] THEN
        APPEND alice_bits[i] TO shared_key
    END IF
END FOR

// --- STEP 5: Output ---
PRINT "Alice bits: ", alice_bits
PRINT "Alice bases: ", alice_bases
PRINT "Bob bases: ", bob_bases
PRINT "Bob results: ", bob_results
PRINT "Shared secret key: ", shared_key
PRINT "Key length: ", LENGTH(shared_key)
PRINT "Eve intercepted: ", LENGTH(eve_indices), "
photons"

END BB84_SIMULATION

```

Logical Flow Explanation

1. Initialization and Input Validation:

The algorithm starts by accepting the number of photons (n) to be simulated. A validation check ensures that n stays within the acceptable range (8–32). If not, it defaults to 8 photons for stable execution.

2. Step 1 – Alice’s Preparation:

Alice generates a random bit sequence and assigns a random basis (Z or X) to each photon.

- **Z basis** corresponds to vertical (|) and horizontal (—) polarization.
 - **X basis** corresponds to diagonal (/) and anti-diagonal (\) polarization.
- Each photon is encoded according to its bit and basis, forming a set of qubits ready for transmission.

3. Step 2 – Eve’s Interception:

Eve (the eavesdropper) intercepts a random subset of photons, typically 25% of the total.

She measures each in a random basis, which **collapses the qubit state** due to the **Heisenberg Uncertainty Principle**. When she resends them, errors are introduced—detectable by Alice and Bob later.

4. Step 3 – Bob’s Measurement:

Bob receives each photon and independently chooses a random basis (Z or X) for measurement.

If his chosen basis matches Alice’s original basis, he measures the correct bit.

If not, the measurement yields a random result due to the nature of quantum superposition.

5. Step 4 – Sifting Process:

After transmission, Alice and Bob publicly announce their chosen bases (not bits).

They keep only the bits for which their bases matched.

This subset forms the **shared secret key**. Any discrepancies indicate the presence of an eavesdropper.

6. Step 5 – Output Generation:

The algorithm prints the details of all sequences—Alice’s bits, Alice’s and

Bob's bases, Bob's measured results, the final shared key, and Eve's interception details.

This provides complete transparency and helps analyze **key length, efficiency, and security**.

Advantages of the Algorithm

- **Simplicity:** The structure of the algorithm closely follows the original BB84 protocol, making it easy to implement and understand.
- **Error Detection:** Incorporates random interception to analyze eavesdropping effects.
- **Scalability:** Works for different photon counts and can be extended for hardware simulations using Qiskit.
- **Transparency:** Prints all intermediate values for debugging and educational clarity.
- **Realism:** Introduces measurable quantum disturbances that reflect actual physics behavior.